

From Community Gardens to Rooftops: Practices, Innovation, and Regulations to Enhance Barcelona's Food System

Diego Maximiliano Macall

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Presentation Outline

- 1 Introduction
- 2 Motivations, Hypothesis, Research Questions and Objectives
- 3 Agronomic and Organizational Aspects of Barcelona's Community Gardens
- 4 A food system innovation: Vegetable production in rooftop greenhouses in Barcelona
- 5 Analysis of Regulations Governing the Scale Up of High Technology Urban Rooftop Horticulture in Barcelona
- 6 Thesis Discussion
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Introduction

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 - 2 actively shape its transformation through innovation and policy interventions.

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 - 3 the interactions between these.

- Targeted policy interventions can be used to reintegrate food production into the urban fabric, and guide food systems toward sustainability (Wensing et al., 2023).

Urban Agriculture

- Targeted policy interventions can be used to reintegrate food production into the urban fabric, and guide food systems toward sustainability (Wensing et al., 2023).
- Cities and agriculture co-evolved over 12,000 years, but the last century has witnessed a stark separation between them (Pothukuchi and Kaufman, 1999; Zeunert, 2018).

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- Urban agent *chronesthesia* (time-consciousness).

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- Cities are critical arenas where food governance dynamics play out most visibly (Haysom, 2015; Steel, 2013).
- Municipal governments around the world are not passive. They have increasingly positioned themselves as active players in food system transformation.

Barcelona as a Laboratory for Urban Agriculture: Context, Constraints, and Governance.

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Barcelona in Context Context

Table 1: Comparative Characteristics of Cities at a Similar Latitude

City	Barcelona	Chicago	Rome	Tashkent
Latitude	41°38'N	41°82'N	41°90'N	41°31'N
2025 Population (Millions)	1.6	2.7	2.75	3
Area (km ²)	101	600	1,285	438
Population Density (Inhabitants/km ²)	16,144	4,166	2,220	6,942
Green Space per Capita (m ² /Inhabitant)	7	14.2	39	10*
Annual Sunshine Hours	2,683	2,508	2,473	2,820
Annual Mean Total Precipitation (mm)	582	937	804	419
Participation in the MUFPP	Yes	Yes	Yes	No
Dedicated Urban Agriculture Policies	Yes	Yes	Yes	Yes

Greenery Density in Barcelona Districts

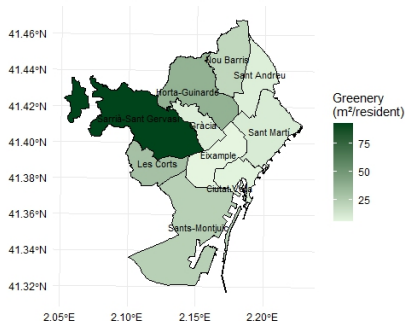


Figure 1: Greenery density (m^2 per resident) across Barcelona districts.

Motivations, Hypothesis, Research Questions and Objectives

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- 2 What proportion of Barcelona's most demanded vegetables could technically be produced on the city's available rooftop area using high technology UH?

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- 2 What proportion of Barcelona's most demanded vegetables could technically be produced on the city's available rooftop area using high technology UH?
- 3 What do Barcelona's existing legal and regulatory frameworks entail for rooftop horticulture, and who ultimately holds the key to the expansion of high technology UH across the city?

Agronomic and Organizational Aspects of Barcelona's Community Gardens

Agronomic and Organizational Aspects of Barcelona's Community Gardens

This chapter is based on the peer-reviewed article: *Macall, D. M., Gabarrell Durany, X., & Villamayor-Tomas, S. (2025). Agronomic and organizational aspects of Barcelona's community gardens. Frontiers in Sustainable Food Systems, 9, 1459265. DOI: <https://doi.org/10.3389/fsufs.2025.1459265>*

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- Consequently, low-tech UH systems in CGs remain the most widespread form of UA globally, owing to their accessibility and lower resource requirements (WinklerPrins, 2017).

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- UA presents a promising approach to produce food within cities.
- Consequently, low-tech UH systems in CGs remain the most widespread form of UA globally, owing to their accessibility and lower resource requirements (WinklerPrins, 2017).
- CGs can be described as “safe havens that provide residents with a sense of nature, community, rootedness, and power” (Schmelzkopf, 1995, p. 364).

Introduction

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- This study examines the demographics, organizational structures, and agronomic practices of CGs within Barcelona's municipal boundaries.

Materials and Methods

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- Ethics approval for this study was obtained from the Comisión de Ética en la Experimentación Animal y Humana of the Autonomous University of Barcelona on July 15, 2022.
- Specialized assessment tool used to visually document five key physical characteristics of the CGs visited during the study.

- A total of 18 gardens, with varying stated main purposes or objectives were visited between July and November 2022.

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 - Community Garden History
 - Garden Decision-Making Mechanism, Member Aspects, and Communication
 - Agronomic Aspects of CGs Visited

Visited Community Garden Typology (I)

Community garden type & interviews completed	Garden code	Garden's stated purpose	Garden size (m ²)	Garden shape	CG decision-making mechanism
Community (2)					
	COM1	Nature in urban area	1,088	Rectangle	Neighbor managed
	COM2	Social cohesion & therapy	422	Square	Administrator takes major decisions
	COM3	Youth centre	456	Irregular	Administrator takes major decisions
Empty Urban Spaces with Territorial and Social Involvement (1)					
	EUSTS1	Temporary administration of unused municipal land	101	Triangular	Decisions left to CG members
	EUSTS2		662	Rectangle	
Civic Centre (1)					

Visited Community Garden Typology (II)

Community garden type & interviews completed	Garden code	Garden's stated purpose	Garden size (m ²)	Garden shape	CG decision-making mechanism
Barcelona Network of Urban Gardens (9)					
	BNUG1	≥ 65-year-old Barcelona citizen involvement in UA	369	D-shaped	Municipality sets rules; CG administrator enforces
	BNUG2		5,620	L-shaped	
	BNUG3		2,807	Square	
	BNUG4		532	Triangle	
	BNUG5		736	Square	
	BNUG6		667	Rectangle	
	BNUG7		1,462	Rectangle	
	BNUG8		10,115	Rectangle	
	BNUG9		1,111	Triangle	

Results



BNUG 2 (A)

BNUG 2 (B)

Figure 2: Polyculture in Barcelona's CGs.

Results

Table 2: Crops grown in Barcelona's community gardens

Common name	Latin name	Conventional yield (kg/m ²)	Organic yield (kg/m ²)
Basil	<i>Ocimum basilicum</i>	360 leaves per plant	219 leaves per plant
Bell pepper	<i>Capsicum annuum</i>	22.5	15.3
Cucumber	<i>Cucumis sativus</i>	27.5	18.7
Eggplant	<i>Solanum melongena</i>	23.7	16.6
Lettuce	<i>Lactuca sativa</i>	22.6	15.4
Onion	<i>Allium cepa</i>	31.3	21.3
Pumpkin	<i>Cucurbita maxima</i>	35.6	24.2
Squash	<i>Cucurbita</i>	29.1	19.8
Tomato	<i>Solanum lycopersicum</i>	33.0	22.4
Zucchini	<i>Cucurbita pepo</i>	29.5	20.0

Results



CIV 1

BNUG 8

BNUG 4

Figure 3: Elevated beds or Tables for Horticultural Production.

The Need for Municipal Coordination and Professionalization

- Achieving significant production levels would require the adoption of monoculture practices for specific crops currently cultivated in these gardens.

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- Realizing this goal necessitates greater involvement and coordination by municipal authorities to streamline production efforts.

A food system innovation: Vegetable
production in rooftop greenhouses in Barcelona

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- Historically, the understanding of food systems has been shaped by rural and agricultural perspectives.
- As cities become increasingly important nodes within global food systems, integrating food production into urban infrastructure requires addressing multiple constraints, including space availability and economic viability.
 - While cities typically lack the productive components of their respective food system, it is feasible to integrate agricultural production into city infrastructure (Smit et al., 1996).

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- This study empirically measures the production potential of three fresh consumed vegetables in Barcelona within RTGs.

Materials and Methods

- Tomatoes, bell peppers, and lettuce are three of the twenty-one most consumed vegetables by Catalanian citizens according to the Spanish Ministry of Agriculture, Fisheries and Food's Household Food Consumption Panel (MAPA, 2024).

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- Together, these three vegetables represented 31% of total Barcelonan per capita consumption of fresh (not cooked or processed) vegetables in 2022.

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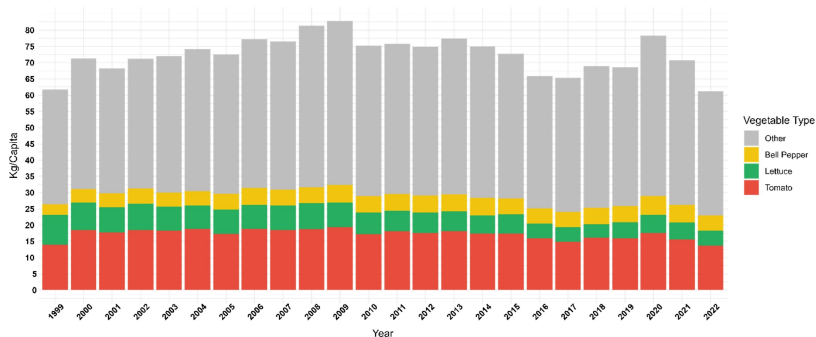


Figure 4: Catalanian annual per capita tomato, bell pepper, and lettuce consumption 1999-2022.

Source: Author based on MAPA (2024).

Materials and Methods

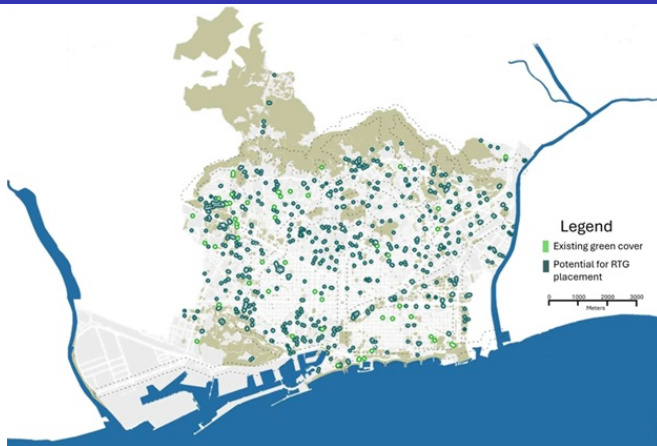


Figure 5: Spatial distribution of potential RTG sites and existing green cover in Barcelona.

Source: Adapted from AEUB (2010, p. 180).

RTG Establishment and Crop Allocation

$$\max R = 9.82 x_1 + 3.47 x_2 + 0.28 x_3$$

$$\text{s.t. } x_1 + x_2 + x_3 \leq 650,000 \quad (1)$$

$$x_1 = 390,000 \quad (2)$$

$$x_1, x_2, x_3 \geq 0$$

RTG Tomato, Bell Pepper, and Lettuce Yield

- Crop trials undertaken by Rufí-Salís et al. (2020)

RTG Tomato, Bell Pepper, and Lettuce Yield

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- Monte Carlo method as specified by Oberle (2015)
 - 960,400 simulations

RTG Tomato, Bell Pepper, and Lettuce Yield

Table 3: Maximum and minimum crop yields within RTGs.

Crop	Season	Max yield (g/plant)	Min yield (g/plant)	MC simulated yield per plant (g/plant)	Yield (kg/m ²)
Tomato	Wi-Sp-Su	8,463	5,179	6,821	13.6
Bell Pepper	Au-Wi	3,150	750	1,949	3.9
Lettuce	Sp	379	174	277	0.6

Water Requirements

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 - 5,878 t requires 423,216 m³ of water to produce
 - 507 t requires 63,882 m³ of water to produce
 - 78 t requires 4,212 m³ of water to produce

Results



Figure 6: Average total tomato, bell pepper, and lettuce consumption between 1999-2022 in contrast to total production of these crops within 65 ha of RTGs

- Through the scaling-up of UH within RTGs to a city-wide level, it is possible to supply a considerable amount of the tomatoes, bell peppers, and lettuce consumed by Barcelonans.

Production Sensitivity Analysis

Table 4: Total tomato, bell pepper, and lettuce production attainable through 90 m² RTGs.

Crop	Yield (kg/m ²)	Total buildings	Production per single RTG unit	Total production (t)
Tomato	13.6	437	1,224	535
Bell Pepper	3.9	437	351	153
Lettuce	0.6	437	54	24

Conclusion

- The substantial production of vegetables that can be achieved in a mere 65 ha of rooftop area is informative for Barcelona's municipal authorities, as they consider current and future food and agricultural policies.

Analysis of Regulations Governing the Scale Up of High Technology Urban Rooftop Horticulture in Barcelona

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- Municipal authorities have the ability, through targeted policy interventions, to reintegrate food production into the urban fabric and guide food systems toward sustainability (Wensing et al., 2023).

Barcelona as a Case Study

- Unique legal status within Spain, as outlined in *Article 89 of the Statute of Autonomy of Catalonia* (Catalunya, 2006).

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- This study identifies the laws and ordinances regulating rooftop use in Barcelona and analyzes the institutional statements within these them that either directly or indirectly facilitate or restrict the installation of high technology UH.

Key institutions governing Barcelona's rooftop area

Table 5: Key institutions governing Barcelona's rooftop area.

Institution	Geographic Scope	Effective Since	Aim	Article(s)
Spanish Civil Code (Law/Royal Decree)	Spain	July 24, 1889	Property rights	396
Horizontal Property Law (Law 49/1960)	Spain	July 23, 1960	Multiple owner building management	17.4
Codi Civil de Catalunya (Civil Code of Catalonia)	Catalonia	Feb 6, 2003	Multiple owner building management	553-26(c), 553-43
Ordinance on the Uses of the Urban Landscape of the City of Barcelona	Barcelona Metropolitan Area	July 19, 1999	Regulates private use that distort historic, artistic, or traditional ensembles	74, 76

Source: Ajuntament de Barcelona (2024).

Institutional Grammar characteristics used in this study

Table 6: Institutional Grammar characteristics used in this study

Component	Acronym	Specification
Attribute	A	The individual(s), group(s), or organization(s) that carry out the alm.
oBject	B	Animate or inanimate part of the statement that is the receiver of an action.
Deontic	D	The prescriptive component of an institutional statement that specifies what is allowed, required, or prohibited.
alm	I	The alm specifies the objective or action described by the institutional statement to which the Deontic refers.
Condition	C	Part of the statement that modifies the alm (in temporal or spatial terms) or describes how the action recognized in the alm is to occur.
Or else	O	Punitive action if the rule is not adhered to.

Source: Based on Crawford and Ostrom (1995), Ostrom (2005), and Siddiki et al. (2011).

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 - 3 Analyzing Institutional Statements Using the IGT Framework.

Table 7: Summary statistics of policy instruments deconstructed

Legal instrument	Num of articles dissected	Num of institutional statements extracted
Horizontal Property (Law 49/1960)	1	19
Civil Code of Catalonia	2	14
Ordinance on the uses of the urban landscape of the city of Barcelona (Norm)	2	64

Results

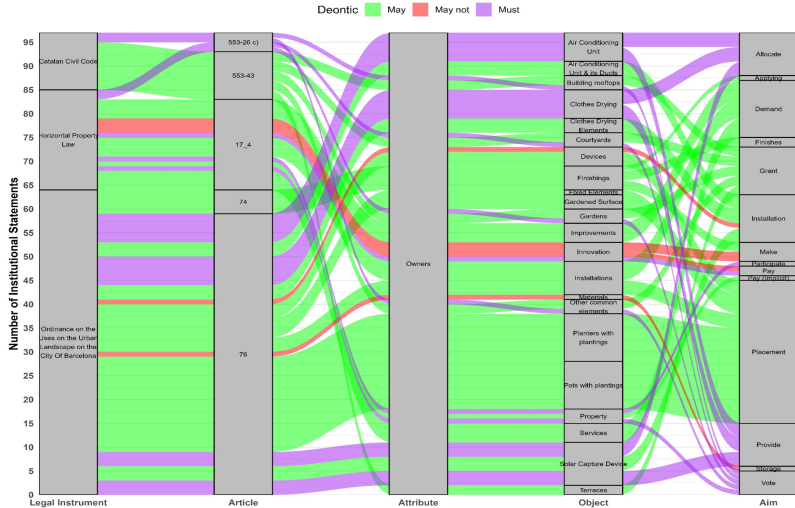


Figure 7: Dissection of Policy Instruments and relevant Articles to the integration of UH onto Barcelona's buildings

- A novel employment of the IGT

Discussion

- A novel employment of the IGT
- Stakeholder Semi-Structured Interviews

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- However, its scalability ultimately depends on building owners, as they hold the power to enact modifications to the rooftops of their properties.

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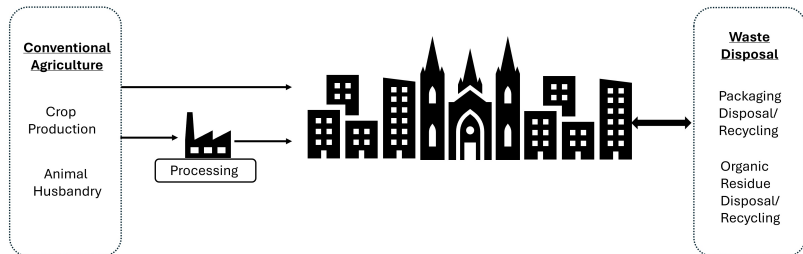


Figure 8: Dissection of Policy Instruments and relevant Articles to the integration of UH onto Barcelona's buildings

Thesis Discussion

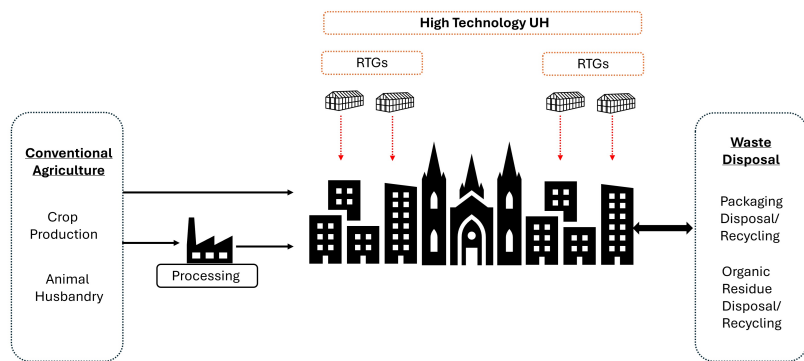


Figure 9: Dissection of Policy Instruments and relevant Articles to the integration of UH onto Barcelona's buildings

Thesis Discussion

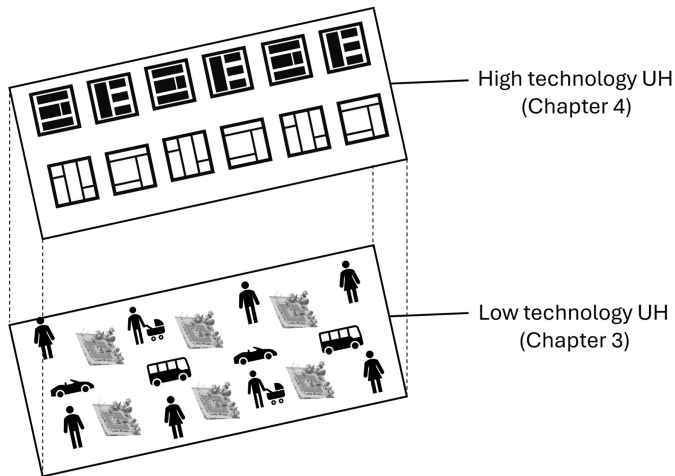


Figure 10: Dimensional Analysis of UH. The top panel shows the city's rooftop area, and the bottom one shows the ground level area where UH could be located.

Thesis Conclusion

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- As the 21st century unfolds, humanity faces a set of unprecedented challenges, many of which converge in cities, the principal arenas of modern human activity.
- In response to mounting pressures on its food system, Barcelona's municipal authorities have articulated a forward-looking vision: reintroducing food production into the urban fabric to, among other things, make their food system more resilient. This dissertation tests the practical potential of UH in Barcelona at both the ground and rooftop levels.

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- Dr. Gabarrell & Dr. Villamayor

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- Dr. Gabarrell & Dr. Villamayor
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